

Course Syllabus

General Information

Faculty	Information Technology		
Department	Computer Science		
Semester	First	Academic Year	2025/2026

Course Details

Course Title	Database Systems			Course No.		0413403	
Type of Learning	☒ On- campus ☐ Blended (..... Platform) ☐ Online (..... Platform)						
Credit Hours	2	Theoretical	2	Practical	1		
Course Level (according to JNQF)			7	JNQF Hours*		16.2	
Pre-requisite	System Analysis and Design			Co-requisite		0412401	

Course Instructor Information

Instructor Name	Prof. Mohammed Otair		
Instructor E-mail	m.otair@meu.edu.jo		
Course Time			
Office Hours			
Office Number	N116	Office Phone	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

Fundamentals of Database Systems, R. Elmasri and S. Navathe, Published by Pearson (July 14th 2021)

2. References

- Database Systems: Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, 7th Edition
- Database Design for Mere Mortals: A Hands-On Guide to Relational Database Design – 2nd Edition, 2013.

Teaching Methods

1. Lectures
2. Discussion and problem solving
3. Individual assignments
4. Groups discussion
5. Case study

Course Description

This course introduces the fundamental principles of relational database design and management: Relational Database Management Systems; Entity Relationship Modeling; Structured Query Language (SQL); Database schema: design normalization, redundancy elimination.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Define the basic data management concepts.	PLO1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	X		
2	Design a database system using Entity-Relationship Diagram (ERD) and Enhanced Entity-Relationship Diagram (EERD).	POL2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.		X	
3	Describe the basic concepts of the relational data model; tables and relations	PLO1: Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	X		
4	Use Structured Query Language (SQL) to create, manipulate and query a database.	POL2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.		X	
5	Apply the normalization process to relational databases.	POL2: Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.		X	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative / Formative
1	Introduction to Databases Lab 1: Introduction	Lecture (1)	Lectures	1	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Lectures	1	Quiz#1, Mid exam, final exam	Summative
		Lab 1	Lectures	1	Discussion & Question	Formative
2	Database System Concepts Lab 2: Create tables	Lecture (1)	Lectures	1	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Lectures	1	Quiz#1, Mid exam, final exam	Summative
		Lab2	Lectures	2	Discussion & Question	Formative

3	Database System Architecture Lab 3 :Delete tables	Lecture (1)	Lectures	1	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Lectures	1	Quiz#1, Mid exam, final exam	Summative
		Lab3	Lectures	2	Discussion & Question	Formative
4	Data Modeling Using Entity-Relationship (ER) Model Lab 4: Alter table and keys	Lecture (1)	Lectures	2	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Case Study	2	Quiz#1, Mid exam, final exam	Summative
		Lab4	Lectures	2	Assignment	Summative
5	Enhanced Entity-Relationship (EER) Modeling Lab 5: keys & DML Statements	Lecture (1)	Lectures	2	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Case Study	2	Quiz#1, Mid exam, final exam	Summative
		Lab 5	Lectures	2	Discussion & Question	Formative
6	Relational Database Design (Mapping) Lab 6: Foreign Keys	Lecture (1)	Lectures	3	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Lectures	3	Quiz#1, Mid exam, final exam	Summative
		Lab6	Lectures	3	Assignment	Summative
7	Relational Database Constraints Lab 7: Select Statement Part1	Lecture (1)	Lectures	3	Quiz#1, Mid exam, final exam	Summative
		Lecture (2)	Lectures	3	Quiz#1, Mid exam, final exam	Summative
		Lab7	Lectures	4	Discussion & Question	Formative
8	Mid exam					
9	Relational DB Design by ER and EER-to-Relational Mapping Lab 8: Select Statement Part2	Lecture (1)	Interactive	4	Quiz#2, Mid exam, final exam	Summative
		Lecture (2)	Interactive	4	Quiz#2, final exam	Summative
		Lab8	Lectures	4	Discussion & Question	Formative
10	Basic SQL Lab 9: Create users	Lecture (1)	Interactive	4	Quiz#2, final exam	Summative
		Lecture (2)	Interactive	4	Quiz#2, final exam	Summative
		Lab9	Lectures	4	Discussion & Question	Formative
11	Complex SQL Lab 10: grant privilege for users	Lecture (1)	Interactive	4	Quiz#2, Mid exam, final exam	Summative
		Lecture (2)	Interactive	4	Quiz#2, final exam	Summative
		Lab10	Lectures	4	Assignment	Summative
12	Relational Algebra and Calculus	Lecture (1)	Lecture	5	Quiz#2, final exam	Summative
		Lecture (2)	Groups discussion	5	final exam	Summative
		Lab11	Lectures	4	Discussion & Question	Formative
13	Basics of Functional Dependencies	Lecture (1)	Lecture	5	final exam	Summative
		Lecture (2)	Groups discussion	5	final exam	Summative
		Lab12	Lectures	5	Assignment	Summative
14	Normalization for Relational Databases	Lecture (1)	Lecture	5	final exam	Summative
		Lecture (2)	Lecture	5	final exam	Summative
		Lab13	Lectures	5	Discussion & Question	Formative
15	Project Presentation	Lecture (1)	Groups discussion	5	Presentation	Summative
		Lecture (2)	Groups discussion	5	presentation	Summative

16	Final Exam			1-5		Summative
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Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
Written Exam/ Mid Term Exam	30%	Week 8
Written Exam/ Final Exam	40%	Week 16
Quizzes	-	
Assignment	20%	Week 4,6,11,13
Project	10%	Week 15

Course Policies

Item	Policy
Quizzes	
Exams	The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	Makeup exam should not be given unless there is a valid excuse.
Drop Date	Last day to drop the course is according to the University regulations.
Academic Honesty	- Cheating or copying on exam or quiz is an illegal and unethical activity. - Standard MEU policy will be applied. - All graded assignments must be your own work (your own words).
Attendance	- Excellent attendance is expected. - MEU policy requires the faculty member to assign ZERO grade if a student misses 20% of the classes. - Sign-in sheets will be circulated.
E-Learning	It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	The student should expect to spend 6 hours per week as an average workload.
Graded Exams	Instructor should return exam papers graded to students within one week after the exam date.
Participation	- Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any question. - Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	- Homework's are as mentioned in the previous table. - Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. - More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (If Applicable)

Criteria	Excellent (5)	Very Good (4)	Adequate (3)	Poor (2)
Create ER and EER Diagrams	Diagrams are comprehensive, accurately depicting all entities, relationships, and attributes; uses standard notation effectively.	Diagrams are clear with minor omissions; most entities and relationships are represented accurately.	Diagrams are basic with some inaccuracies; missing some key entities or relationships.	Diagrams are unclear or poorly constructed; many essential elements are missing.
Map ER Diagrams to Tables	Accurate mapping of all entities to tables; identifies primary and foreign keys correctly; relationships are well-defined.	Mostly accurate mapping; minor errors in key identification; most relationships are correct.	Some correct mapping; several errors in key identification or relationships present.	Poor or incorrect mapping; significant elements do not translate well to tables.
Create and Populate Tables	Tables are efficiently created with appropriate data types; data is inserted accurately and completely, showing a good understanding of normalization.	Tables are mostly correct with minor data type issues; some entries may be missing or incomplete.	Tables created but with several issues; numerous data entries are incorrect or missing.	Tables are poorly constructed; data insertion is largely incorrect or incomplete.
Write and Execute SQL Select Statements	Select statements are well-formed, retrieving accurate and relevant data; demonstrates understanding of complex queries (e.g., joins, subqueries).	Mostly correct select statements; retrieves data but may lack complexity or contain minor errors.	Basic select statements that retrieve limited data; several inaccuracies present.	Select statements are mostly incorrect; fails to retrieve relevant data effectively.

*** JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	2	15	30
Tutorials			
laboratory	2	15	30
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	2
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	0	0	0
Self-Learning Activities			
Assignments/ Homework	4	4	16
Case Studies	4	2	8
Presentations	1	2	2
E-learning			
Extra Readings	6	12	72
Total Hours	162		
JNQF Hours	=162/10 =16.2		